Exploring eDNA in SBDI

BioSyst.EU 2026

SBDI eDNA team:

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What is SBDI ?

Swedish Biodiversity Data Infrastructure

- National research infrastructure supported by the Swedish Research Council (VR)
- 11 universities and government agencies
- Host: Swedish Museum of Natural History
- **Swedish GBIF node**: national access point for **Open biodiversity data**:
  - Citizen science (bird watchers, amateur botanists...)
  - Systematic monitoring programs
  - Natural history collections
  - Research projects, including eDNA data
- **Tools and services** for interdisciplinary research on biodiversity and ecosystems

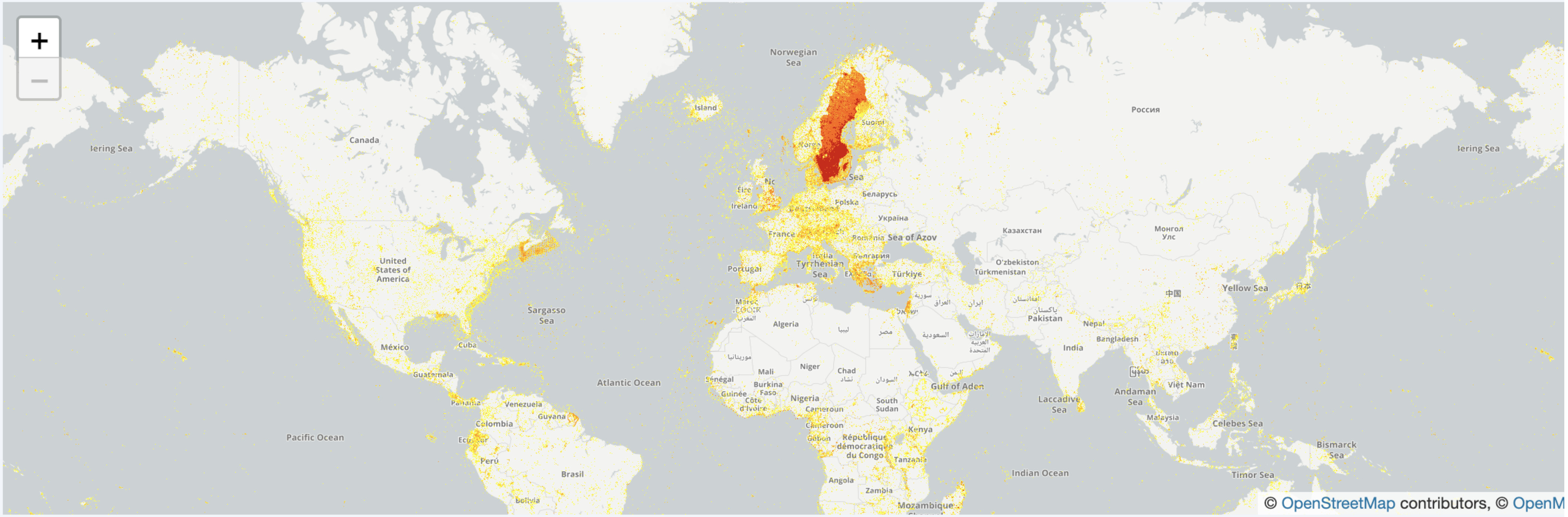
GÖTEBORGS UNIVERSITET · Karolinska Institutet · KTH · Linnaeus University · LUND UNIVERSITY · Naturhistoriska riksmuseet · SLU · SMHI · Stockholm University · UMEÅ UNIVERSITET · UPPSALA UNIVERSITET · GBIF · Vetenskapsrådet

We mobilize biodiversity data

gbif.se

Species occurrences published in GBIF by Swedish institutions

175,556,030	213	251	43
Occurrences	Datasets	Countries covered	Publishers



Explore on gbif.org

We provide tools and services

biodiversitydata.se/explore-data/

Swedish Biodiversity Data Infrastructure

HOME | EXPLORE DATA | PUBLISH DATA | EVENTS | SUPPORT | ABOUT

- Explore data
- Tools and data
- Programming and API:s
- Acknowledging SBDI

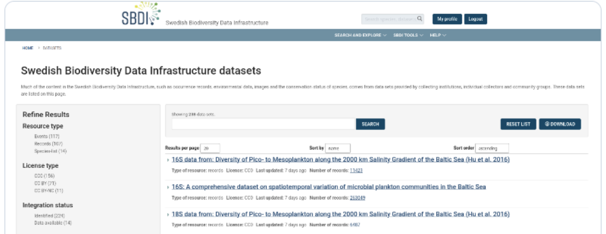
Tools and data

Here we list and describe all SBDI tools and portals that can be used to explore, download, and analyze biodiversity data. Our core tools and data access functionality is based on Living Atlases technology, but we also provide several specialized tools which are also linked to below.

You can filter the list by selecting one of the tags:

Search biodiversity datasets | Spatial analysis | Molecular/genetic data | Systematic inventories | Marine data | Biologging data | Archaeological data | Data flows | Visualize/explore | Project management | Programming

SBDI Bioatlas

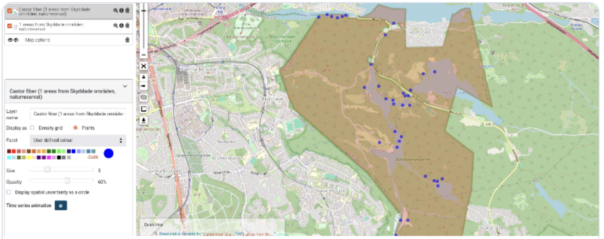


Bioatlas – find and access biodiversity data

The Bioatlas allows you to find and access Swedish biodiversity data, and includes:

- The collectory: browse data providers and datasets
- A search interface: search for data

The SBDI Bioatlas is the Swedish instance of the core components in the Living Atlases platform.

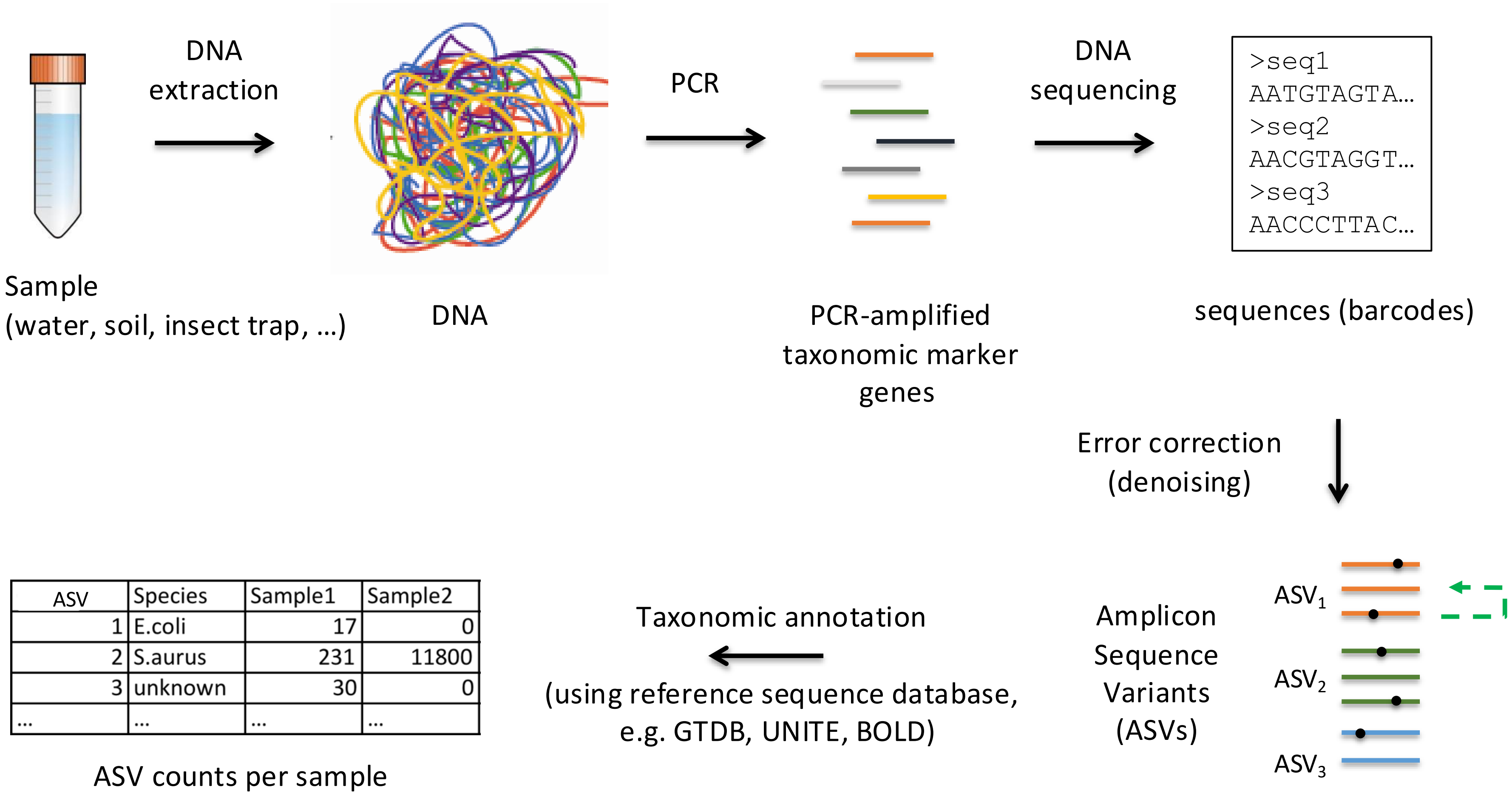


Spatial Portal – visualize and analyze biodiversity data

The Spatial Portal is a tool that allows users to visualize and analyze Swedish biodiversity data in a spatial context – to understand and analyze biodiversity and its relationship with the environment.

- Map and analyze species occurrences alongside environmental and contextual data: map species distributions, explore relationships between species and environment, and conduct spatial analyses

Metabarcoding / eDNA

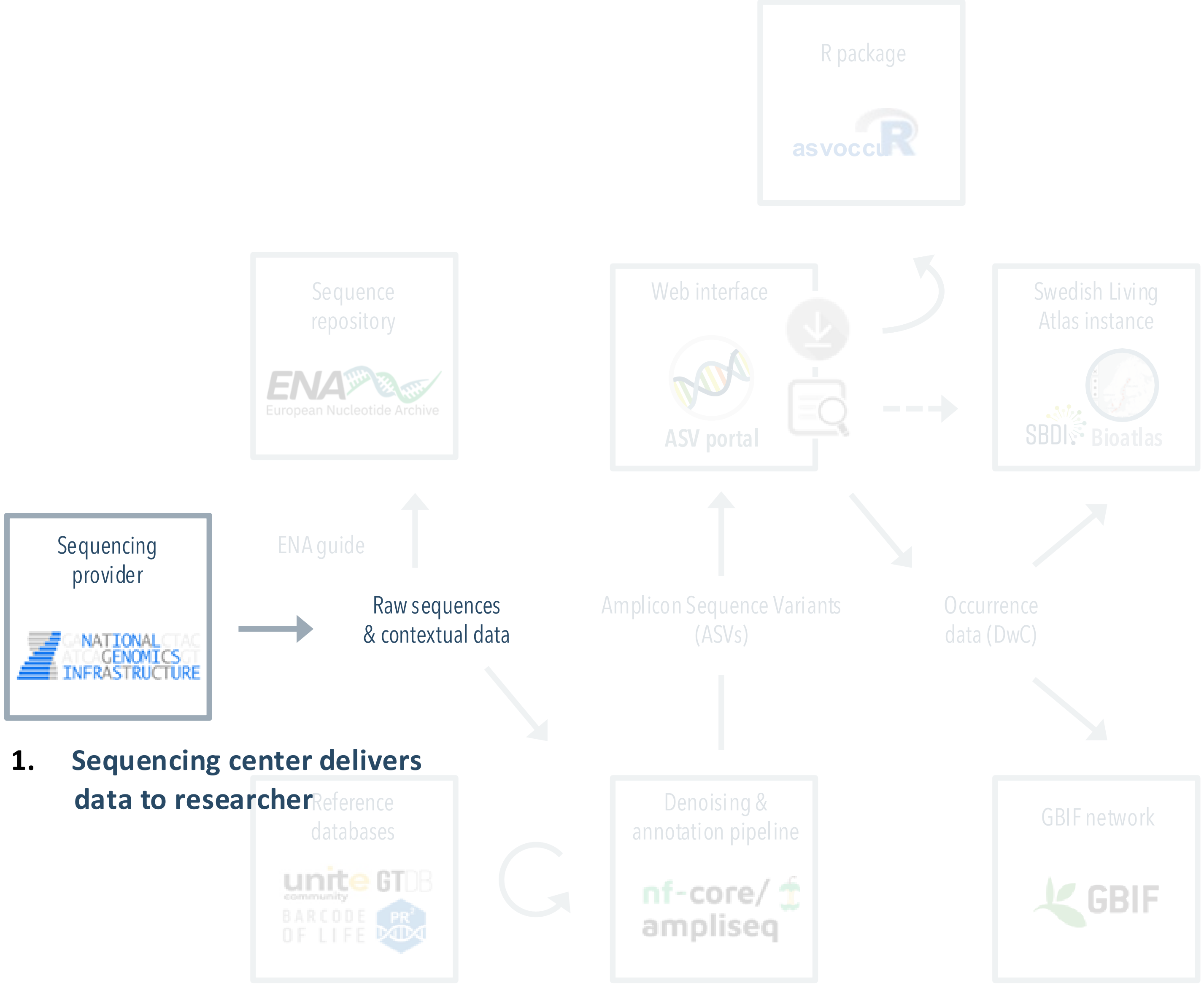


ASV	Species	Sample1	Sample2
1	E.coli	17	0
2	S.aurus	231	11800
3	unknown	30	0
…	…	…	…

ASV counts per sample

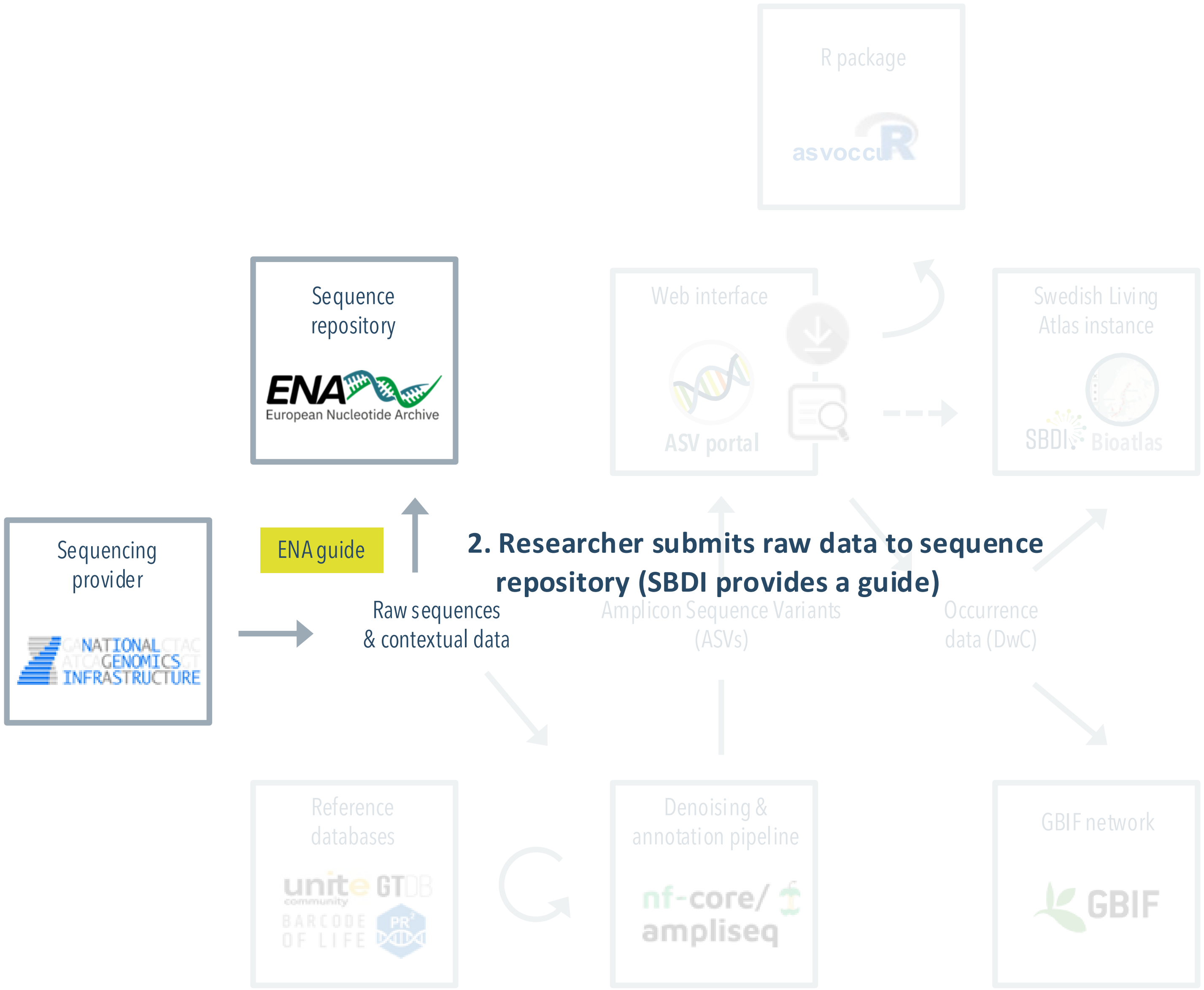
eDNA in SBDI
- tools and data flow...

eDNA in SBDI: tools & data flow



1. **Sequencing center delivers data to researcher**

eDNA in SBDI: tools & data flow



Step 3: Register samples

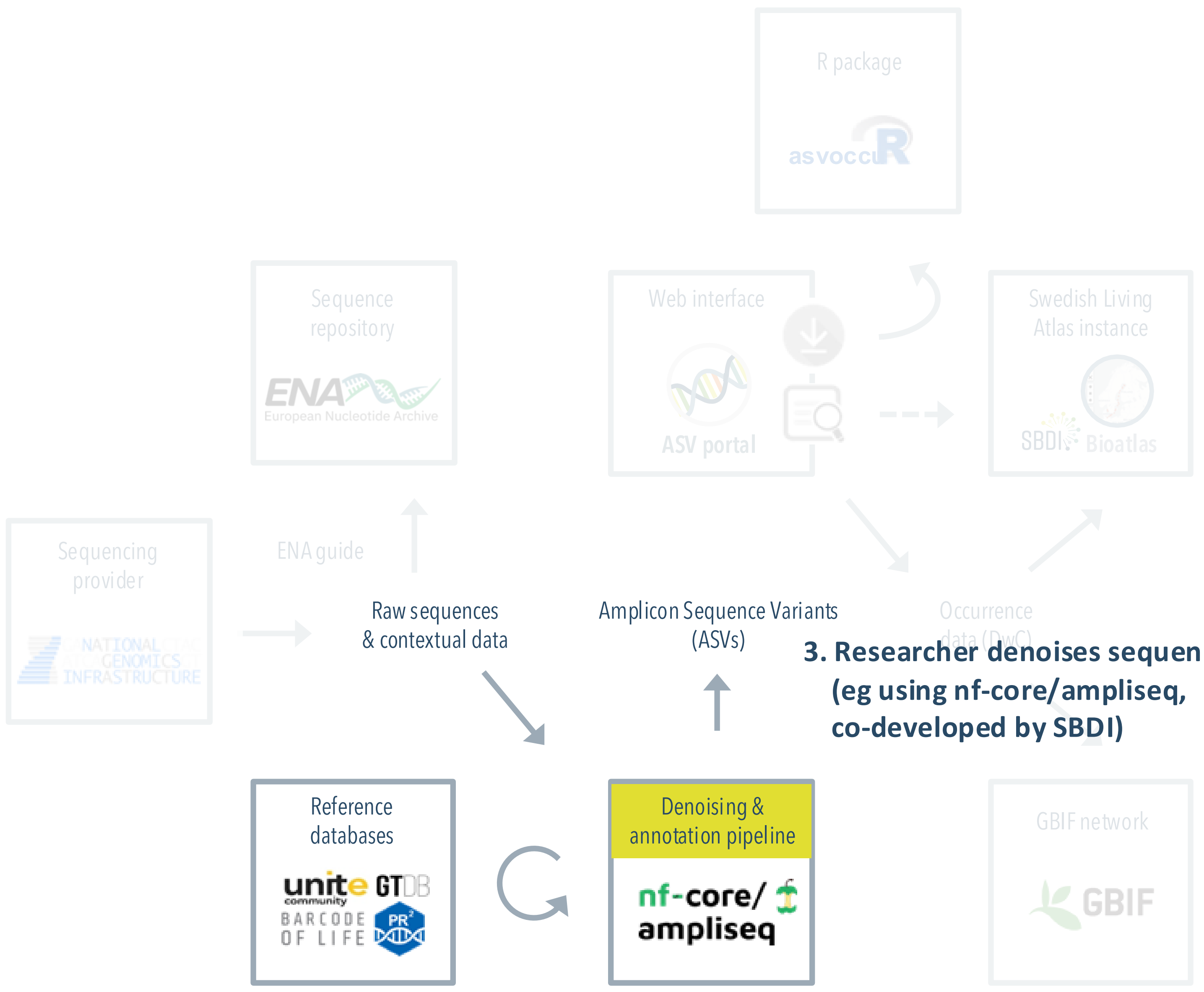
Samples are the source material from which your sequences derive, and the searchability and usability of your submitted data will depend on how well you document these samples. Go to *Samples* | *Register Samples* and click *Download spreadsheet to register samples* to start the process.

Step 3a: Select sample checklist

The ENA sample checklists are partly overlapping sets of attributes (or data fields) that can be used to describe samples, and by selecting one of these you enable your sample metadata to be validated for correctness during submission. For environmental and organismal (host-associated) samples, alike, we recommend using one of the *Environmental Checklists* and, among these, to select the alternative from the *Genomic Standards Consortium (GSC) MIxS checklists* that provides the most specific match to your sampled environment, for example:

Sampled environment	Recommended checklist
Air or general, above-ground, terrestrial	GSC MixS air
Epi- or endophytic (e.g. leaf, root)	GSC MIxS plant associated
Epi- or endozoic (e.g. spider gut, animal skin)	GSC MIxS host associated
Fresh- or seawater	GSC MixS water
Human gut / oral / skin / vaginal	GSC MIxS human gut / oral / skin / vaginal
Human non- gut / oral / skin / vaginal	GSC MIxS human associated
Sediment	GSC MixS sediment
Soil	GSC MixS soil

eDNA in SBDI: tools & data flow



3. Researcher denoises sequences to ASVs (eg using nf-core/ampliseq, co-developed by SBDI)

nf-co.re/ampliseq/2.18.0

Home Pipelines Resources Docs Community About Search Join nf-core

nf-core/ampliseq

Amplicon sequencing analysis workflow using DADA2 and QIIME2

16s 18s amplicon-sequencing edna illumina iontorrent its metabarcoding metagenomics metataxonomics microbiome pacbio qiime2 rrna taxonomic-classification taxonomic-profiling

Launch version 2.18.0

https://github.com/nf-core/ampliseq

Daniel Lundin (LnU)

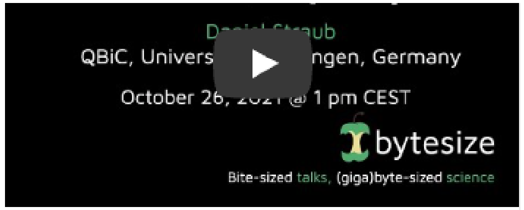
See ampliseq workshop 13-14:30 today

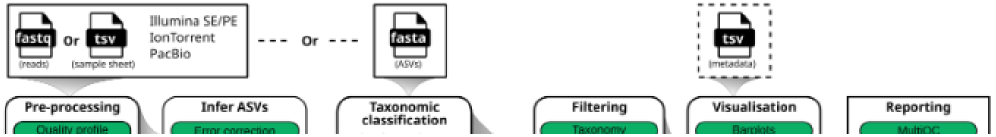
Introduction Usage Parameters Output Results

Introduction

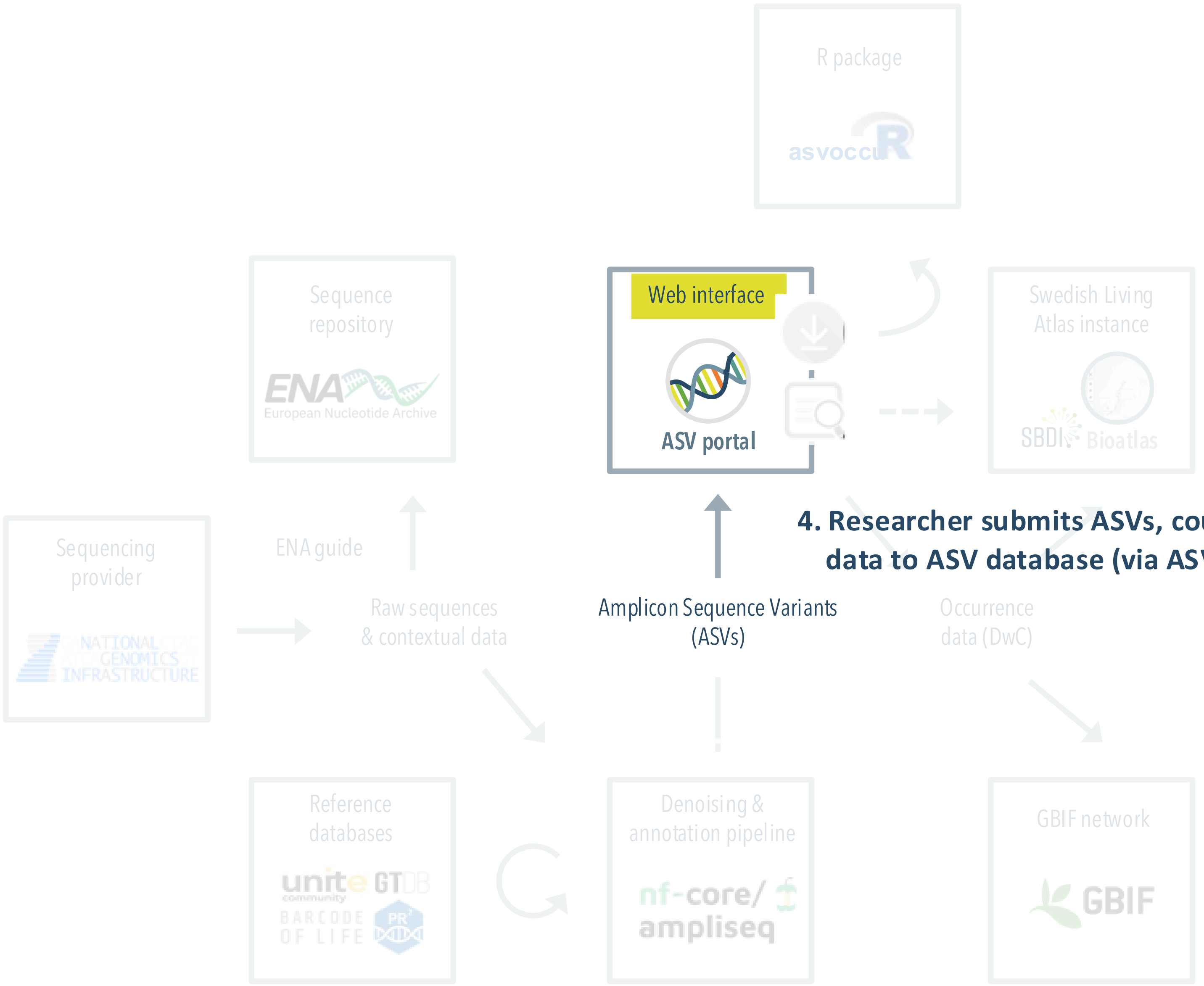
nfcore/ampliseq is a bioinformatics analysis pipeline used for amplicon sequencing, supporting denoising of any amplicon and supports a variety of taxonomic databases for taxonomic assignment including 16S, ITS, CO1 and 18S. Phylogenetic placement is also possible. Multiple region analysis such as 5R is implemented. Supported is paired-end Illumina or single-end Illumina, PacBio and IonTorrent data. Default is the analysis of 16S rRNA gene amplicons sequenced paired-end with Illumina.

A video about relevance, usage and output of the pipeline (version 2.1.0; 26th Oct. 2021) can also be found in YouTube and billibilli, the slides are deposited at figshare.





eDNA in SBDI: tools & data flow



4. Researcher submits ASVs, counts & contextual data to ASV database (via ASV portal)

asv-portal.biodiversitydata.se/index

Swedish Biodiversity Data Infrastructure

Swedish ASV portal

Explore and share DNA sequences and species occurrences derived from metabarcoding (eDNA) studies

Search for ASVs and Bioatlas records using Basic Local Alignment Search Tool (BLAST)

BLAST

Search for ASVs and Bioatlas records using filters on sequencing details and taxonomy

FILTER

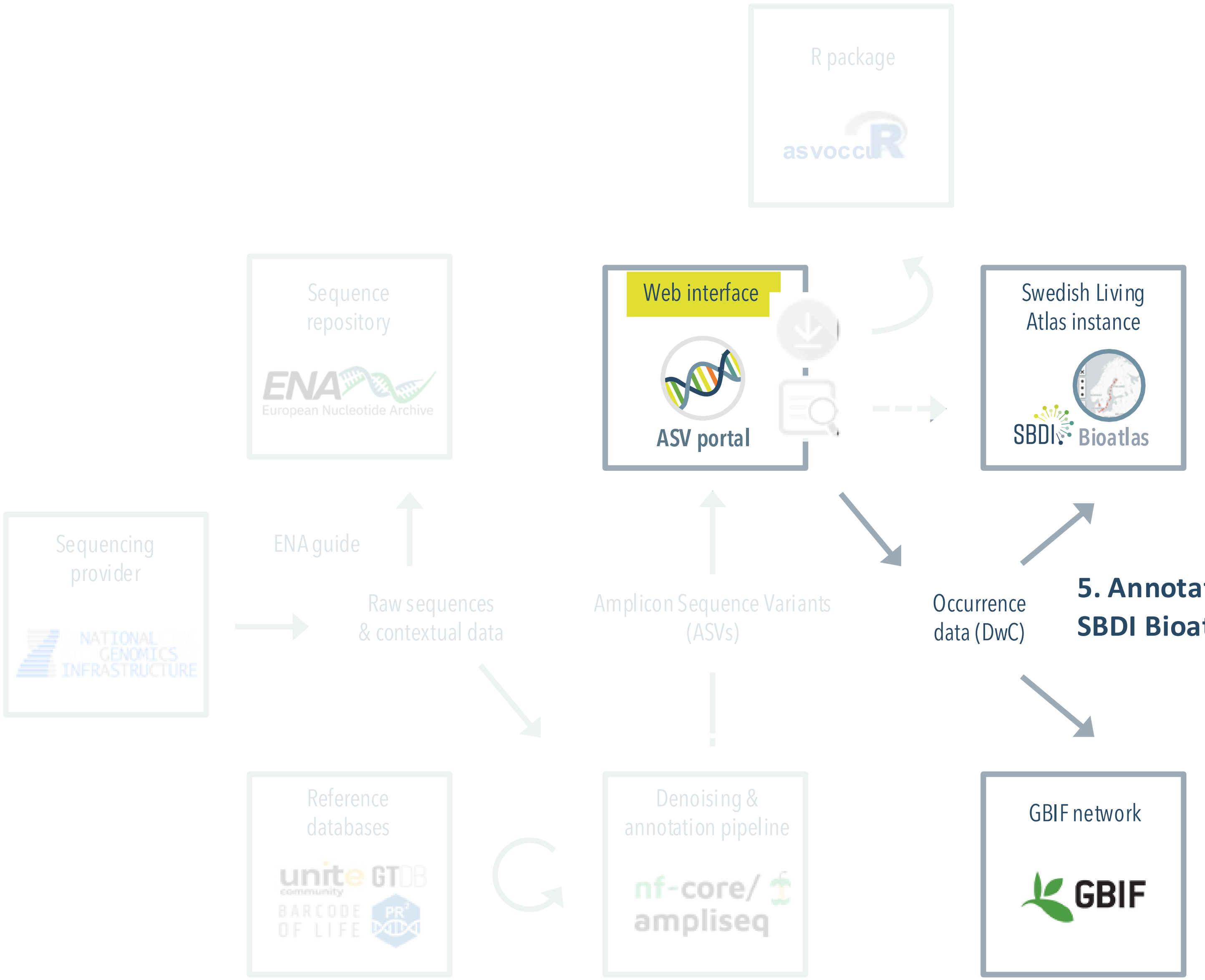
Submit your metabarcoding dataset to the ASV database and SBDI Bioatlas

SUBMIT

Download ASV occurrence datasets, in a condensed Darwin Core-like format

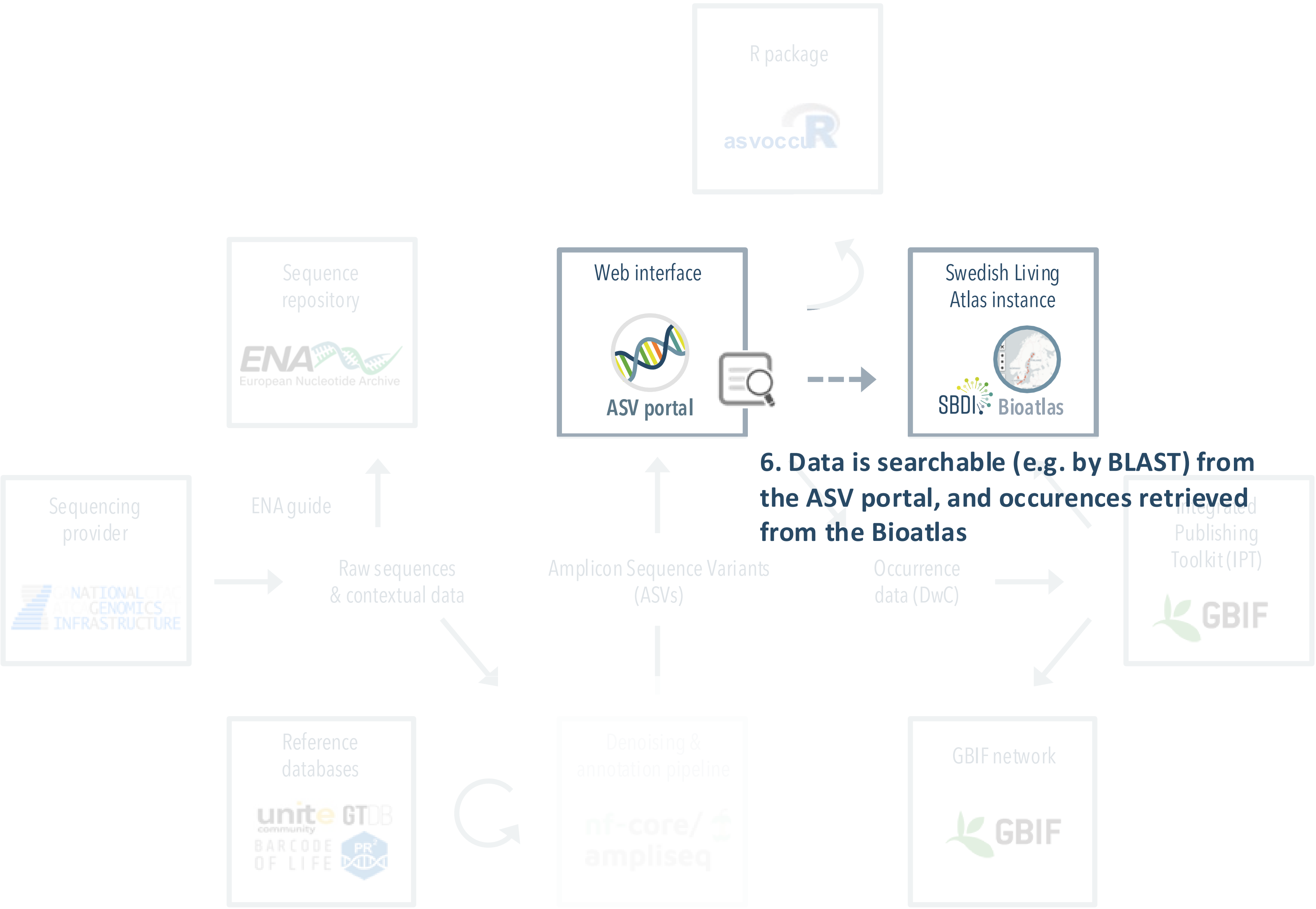
DOWNLOAD

eDNA in SBDI: tools & data flow



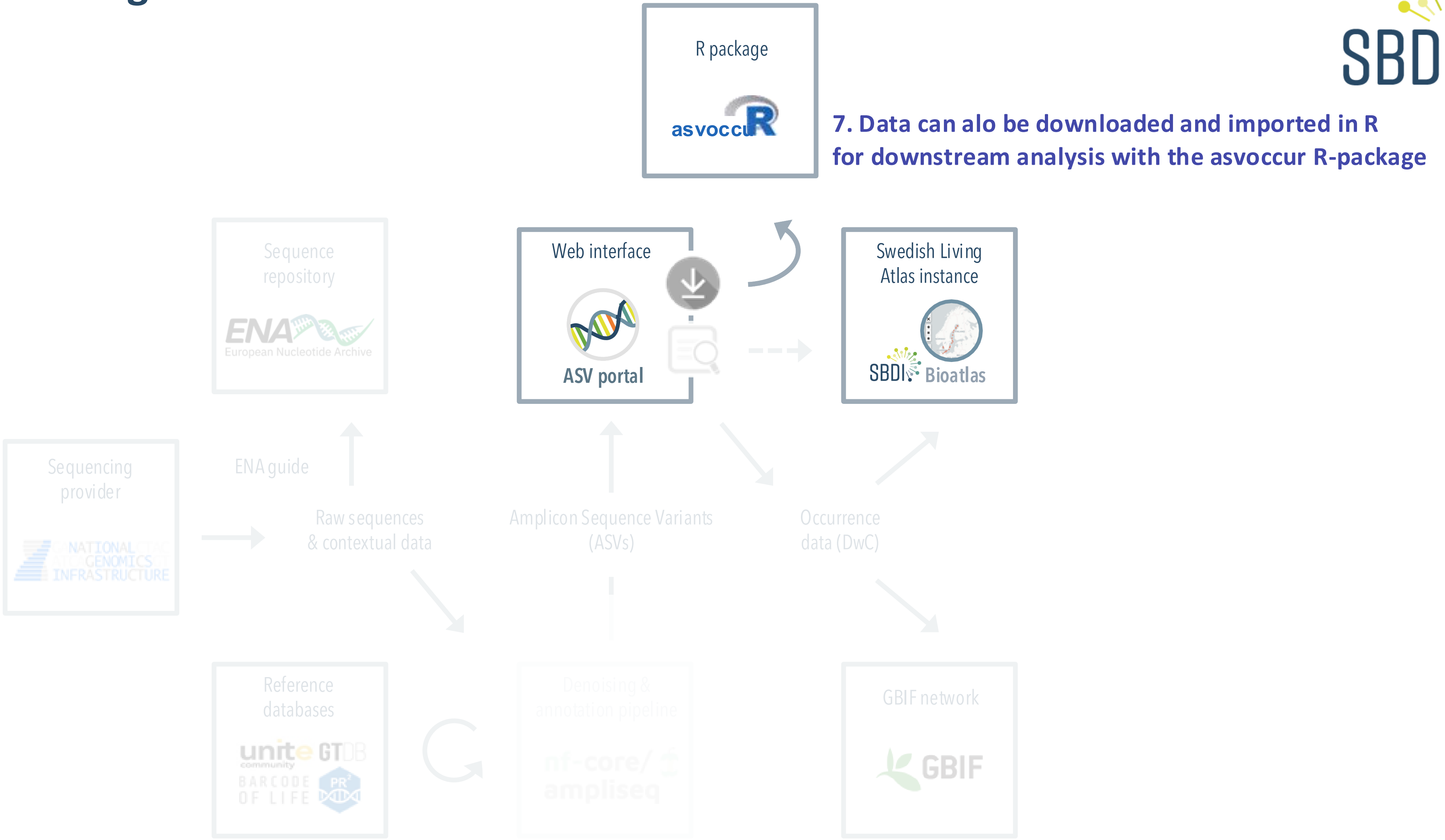
5. Annotated ASVs are integrated into SBDI Bioatlas and (optionally) GBIF

eDNA in SBDI: tools & data flow



6. Data is searchable (e.g. by BLAST) from the ASV portal, and occurences retrieved from the Bioatlas

Metabarcoding data in SBDI: Dataflow



github.com/biodiversitydata-se/asvoccur

README License MIT license

asvoccur

R tools for ASV occurrence data in SBDI.

See Releases for the changelog.

Overview

The **asvoccur** R package, currently under development, provides tools for unpacking and processing ASV occurrence data and metadata downloaded from the Swedish ASV portal. It enables users to convert condensed DwC archives into ASV table format for easier downstream analysis in R, by using functions that load, merge, and aggregate ASV counts across taxonomic ranks.

Tutorial

See our tutorial for an example workflow using **asvoccur** with ASV portal data in R.

Install

```
install.packages('remotes')
remotes::install_github("biodiversitydata-se/asvoccur")
# or:
# remotes::install_github("biodiversitydata-se/asvoccur@develop")
library(asvoccur)
```

Why publish eDNA data in SBDI?

We help make your data FAIR — and keep it valuable over time

- **Standardised** – formatted using Darwin Core to support integration and reuse
- **Citable *** – get a DOI for your dataset
- **Globally accessible *** – contribute to biodiversity knowledge through GBIF
- **Up to date** – benefit from regular ASV reannotation

* If you opt for GBIF export

Tutorial overview

- **Join via** the workshop RStudio Server (grab a tear-off username/password) or use your own laptop

- **Use asvoccur** to load, merge and aggregate Insect Biome Atlas (IBA) data downloaded from the ASV portal

- **Calculate & visualise diversity** across habitat, date and location

Today's tutorial
(& PP slides)

ENA submission guide

Pipeline: nf-core/ampliseq

Web interface: ASV portal

R-package: asvoccur